

Paper 4A Specification v12: Structure

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1. Purpose and position in the programme

Paper 4A is the Structure pillar of the STA measurement work. It estimates the probability that each node in a system is compromised, as a function of structure alone, and tests whether the frozen `cent_core v0.8` model reproduces established regularities of propagation on networks when execution is gated by the node's Cyber Triangle.

The contribution is a confirmatory computational experiment, not a new network theory. Mesh, hub-and-spoke and chain, and the regularities that govern spread on each, are imported from network science, reliability engineering and financial contagion without apology. What is new is the gate: a node can be compromised only where Interface (I), Execution Pathway (X) and Authority (A) all hold, natively or by supply. The paper states how that gate changes the imported regularities and tests whether the model obeys the result. This is the same logic as Paper 3A's Docker correspondence: check the instrument against something known before using it for what is not known.

- Paper 2 (STA) is the theory. 3A is its executable instantiation, frozen at v0.8. 4A is the first confirmatory computational analysis of one dimension and deepens Structure into completion (which triangles close), kind (how they close) and shape (how the closed triangles are arranged). It does not correct Paper 2.
- 4B (Time) and 4C (Adaptation, the game) run on the same saved architectures with Time and Adaptation re-enabled. The pairing of mesh with Structure, hub with Time and chain with Adaptation is stated here as a prediction for them, not tested here.
- Node level (Paper 1A) and relational supply and conferral (Paper 1B) are taken as given. The working hypothesis that the attacker exploits the “or” and the defender wins the “and” belongs to 4C and is not tested here.
- All nodes are treated as equal in value. This is a modelling assumption. Expected compromise volume follows as the sum of node probabilities; node value and the placement of high-value nodes are later work.
- The v8 run (20260919_183415_p4a_all_v8, pooled outcome, citable) is the pilot. Its one supported result, that supplied I, X and A conjunction predicts collapse better than the product of marginal rates, is prior evidence for the gate and is reported as such.

2. Constraints and the record of evidence

Everything new in v12 is driver-level or analysis-level. No mechanic of the frozen core changes.

- One self-contained file, `PhDPaper4A.py`, Python 3.11, run from Visual Studio. No imports of project packages. Third-party libraries: `numpy`, `pandas` and `scipy` only. Graph measures and estimators are implemented in the file.
- `cent_core v0.8` is embedded byte for byte and hash-checked at load against digest `97dbae47`. A mismatch aborts the run. The embedded spec module imports `PyYAML` at module level; if `PyYAML` is absent a stub is installed and the Docker spec files are written by an in-file emitter.
- `VERSION = 12` in the code, this specification and the GitHub tag. The unshipped v9 to v11 drafts are superseded so that one number identifies one design. v8 stays on GitHub as the pilot.
- The run record (`run_record.json`) holds: code SHA-256 with the `RUN_CONFIG` block excluded, this specification's version, core digest and hash-check result, Python and library versions, RNG algorithm (`numpy PCG64`), every factor range, every seed, the phase and shape definitions, and start and end timestamps. `ANALYSE` refuses to produce a citable result if the file's hash differs from the record; `exploratory=True` permits analysis marked not citable.
- Pilot and confirmatory seeds never mix. Pilot seeds 231, 543 and 311. Confirmatory seeds: development 4101, replication 4201 and 4202, held-out 4301. The v5 to v8 pilots informed this design, so their scenarios are excluded from every confirmatory sample.

- Required outputs are never wrapped in try/except. A failure to write any file in Section 16 stops the run. Only optional figures may fail softly, and a failure is logged in the run record.
- Latin hypercube sampling stratifies marginals only. Realised correlations between design factors are computed per sample and reported in exploratory.csv; no correlation control is applied.
- The code is committed and tagged v12 before the confirmatory run starts. The commit timestamp is the evidence that the design preceded the results.
- Every requirement the embedded core cannot meet is listed in Section 17 as a declared model revision candidate, never patched silently.

3. Theory carried into the paper

The paper uses one triangle, three levels of supply, and a strict separation of delivery from execution. These are the definitions every measure and hypothesis relies on.

3.1 One triangle per operation

Every triangle has the same three corners, I, X and A, for one operation on one node. In `cemt_core` v0.8 each node has exactly one specified compromise operation, so node and node-operation are equivalent and completeness is a share of nodes. A node's triangle is complete when every corner holds, either natively (the node holds it) or by supply (a relation stands in for it). A node belongs to many groups (segments, channels, control planes) that can supply corners to many members at once; these are groups, not extra triangles.

3.2 Levels by where the corners sit

The level of a triangle is set by where its corners are supplied from, not by a different set of corners.

Level	Where the corners sit	Incident
Node	all three on the node itself	any single compromise
System	at least one supplied by another node inside the governance boundary	WannaCry lateral spread
System of systems	at least one corner supplied through a relation that crosses the governance boundary	external email or API reach, CrowdStrike update, SolarWinds update

Paper 1B supplies the rule for the third level: every relationally supplied condition is at system-of-systems level when the supplying relation crosses an independent governance boundary. The row does not change. A cross-boundary Connection is a Concentration on the Interface row; a cross-boundary Channel is a Dependency on the Execution Pathway row; a cross-boundary Control Plane is External Trust on the Authority row. External email, web or API reach can therefore be a system-of-systems Interface relation even where execution remains a node-level act initiated by the receiver. Pre-authorisation changes the operational consequence, not the level. "Systemic" is reserved for consequences.

3.3 Relation classes and the layer each occupies

Relation class	Supplies (1B Table 1 row)	Layer	v0.8 delivery rule
Connection	I	standing admission relation from a source at another node	receiver needs its own X and A
Channel	X, assessed conditional on admission; never A	standing route from the receiving side of an Interface to the mechanism	the bundled v0.8 delivery step supplies admission plus this route, so the receiver needs its own A
Conferring control plane	A	privileged identity, outside the network stack	admits nothing; supplies A only

Relation class	Supplies (1B Table 1 row)	Layer	v0.8 delivery rule
Directing control plane	directed exercise of the member's locally held A	designated source whose admitted input is given effect through the local mechanism	v0.8 approximates the composition by supplying I only; the receiver still needs its own X and A

Paper 1B remains authoritative for relation class, row and governance level. Its Channel is an X-row route assessed conditional on admission; it is not redefined here as also supplying Interface. The frozen v0.8 delivery operation (`cemt_core.layers.path_capability`) bundles the admission step with Channel traversal, so a Channel delivery requires only the receiver's Authority. That is an implementation composition of I and X, not a change to Paper 1B's classification. Likewise, a directing Control Plane is an Authority-row relation in Paper 1B, while v0.8 approximates its delivery effect by supplying Interface and requiring the member's X and A locally. The distinction between conceptual classification and frozen implementation is retained throughout.

3.4 Delivery is independent of execution

A payload does not need to compromise the systems that carry it: a mail server relays an infected attachment without running it, a registry serves a poisoned library it never executes, a vendor ships an update whose code runs only on customers' machines. Delivery belongs to the system and system-of-systems layers; execution belongs to the node and is governed by its triangle. Mesh, hub-and-spoke and chain can all deliver inert payloads. In v0.8 only a compromised node passes anything on, so delivery through uncompromised carriers is not represented (Section 17); distance is therefore defined as compromise hops, not network hops.

3.5 Three shapes of triangle fabric

Triangles that share suppliers join into a fabric. Three shapes are the canonical cases, each with an imported law and a defensive lever recorded as a prediction for 4B and 4C.

Shape	Built from	Imported law	Incident	Lever (prediction, not tested here)
Mesh	many connections between neighbours	percolation threshold	WannaCry	Structure: segment
Hub-and-spoke (star)	one control plane or one hub node supplying many	hub fragility (error and attack tolerance)	CrowdStrike	Time: stage the rollout
Chain	single suppliers in sequence	series-system decay with depth	SolarWinds	Adaptation: revoke trust

Real architectures score on all three at once (depth, breadth, redundancy); Section 7 measures them as continuous properties.

3.6 Figures

Three figures are generated from scenario files, not drawn by hand: the eight-glyph key (filled corner held by the node, open corner supplied; native plus seven system kinds), a network of node triangles with dotted supply lines from each supplied corner to the supplier's whole triangle, coloured within versus across the boundary, and the three shapes side by side.

4. What is established, what is built in, what is tested

A reviewer will ask which results the model could have failed to produce. The paper answers by sorting every claim into one of three columns before any run.

Claim	Status	Where it comes from
A node can be compromised only where I, X and A all hold	established	Paper 1A theory; Paper 3A Docker correspondence (tier 1, 20/20 at v0.8)
Relations supply corners; compromise of a supplier confers a position	established	Paper 1B; Paper 3A
Spread can never exceed the set of nodes whose triangle completes (the ceiling)	built in (theorem of v0.8)	cemt_core.layers gating; verified as an implementation check, not a hypothesis
Supplied corners complete triangles	built in	definition; supplied capability beat the marginal product in the v8 pilot, reported as prior evidence
The mesh law: spread switches on past a critical connectivity	tested	percolation theory, imported
The star cut-vertex law: entering at, or removing, the hub has a much larger effect than removing leaves	tested	elementary star topology, motivated by the hub-fragility literature including Albert, Jeong and Barabasi (2000)
The chain law: reach falls geometrically with depth	tested	reliability theory, series systems, imported
The gate shifts each law as site percolation predicts (by the class-specific completion fraction)	tested	the one point of transfer to cyber
The position law: per-node probability falls geometrically with compromise distance, weaker where routes are redundant, and control-plane membership collapses distance	tested	derived in Section 9; Paul's hop intuition
Shape carries information beyond the count of complete triangles	tested	count sufficiency, from the 20 Sept 2026 discussion
Gated distance predicts per-node probability better than ungated distance	tested	topology-only rival, per the v9 review

The computational content of the paper is therefore: how much connectivity it takes to reach the ceiling, the size of each shape's effect, whether the gate shifts each imported regularity by the predicted amount, and whether shape and gated position carry information that counts and ungated topology do not. Those quantities are not

fixed by the model’s definitions. Every citation above is to be verified against the original before the paper cites it.

5. Model configuration on `cemt_core v0.8`

The paper runs on `v0.8` as frozen, in its Structure-only configuration, with every rate fixed. The driver builds scenario specifications and hands them to `build_network` and `run_one_trial` unchanged.

Setting	Value	Reason
Ablation	structure on, time off, adaptation off (Paper 3A’s <code>M_S</code>)	Structure alone; 4B and 4C re-enable Time and Adaptation on the same architectures
Entry	entry pinned to an external source node (native I, X and A, outside the boundary) with <code>entry_certain</code> true; the source is compromised at step 0 and delivers on every relation it supplies	penetration is assumed at the source; compromise is decided node by node by each receiver’s triangle (decision of 20 Sept 2026); no run is lost to an entry gate
Horizon	<code>max_steps</code> 50	frozen default; with Time and Adaptation off the process is susceptible-infected with repeated delivery attempts, so every analytic prediction is stated within this horizon
Gates	<code>authority_rate</code> 0.55, <code>execution_rate</code> 0.55, <code>control_variance</code> 0.0 (the tightest the frozen Beta draw allows, concentration 50)	realised per-node <code>x_prob</code> and <code>a_prob</code> are read from the built network and recorded, so predictions use realised gates
Delivery rates	<code>threat_capability</code> 1.0, <code>beta_conn</code> 0.12, <code>dependency_factor</code> 2.0 (channel base 0.667), <code>control_plane_takeover_rate</code> 0.08, <code>authority_boost</code> 2.0, <code>synchrony</code> 0.5	frozen defaults; per-step delivery probabilities in Section 9 follow from them
Boundary	<code>phantom_cp_rate</code> , <code>phantom_channel_rate</code> , <code>phantom_dep_rate</code> all 0.0	closed boundary; external sources are compromised only through the estate or by pinned entry
Deterministic mode	on for implementation checks and Docker correspondence only	exact reach sets, as in 3A’s structural tier
Node value	impact drawn log-normal (parameters recorded) and reported descriptively only	equal-value assumption for every tested quantity

Completion is class-specific under the frozen delivery rule and the paper uses it that way:

Delivery through	Receiver must hold (natively or by standing supply)	Completion fraction used in predictions
Connection	X and A	<code>f_XA</code>
Channel	A	<code>f_A</code>
Directing control plane push	X and A	<code>f_XA</code>
Source (pinned, entry certain)	nothing; compromised at step 0 by assumption	not applicable

These fractions are computed from the supply table before any run. Standing supply from conferring planes (A), channels (X) and connections (I) counts towards them exactly as `cemt_core.relations` counts it.

6. Architectures

Two families of architecture are built by the driver: canonical shapes, where one thing varies at a time so an imported law can be checked exactly, and mixed architectures, where everything varies so the position law can be tested in the general case. Every architecture is static over its Monte Carlo runs and is saved as a scenario file with its seed for 4B and 4C.

6.1 Canonical shapes (H1, H2, H3, H5)

Common settings: $N = 60$ estate nodes (120 in the held-out sample); $p_I = 0.5$ (native Interface no longer gates delivery from a source, and is kept so Interface measures vary); $p_X = 1.0$; co-location 0. The completion fraction f takes the levels 0.2, 0.4, 0.6, 0.8 and 1.0. Canonical architectures assign exactly $\text{round}(fN)$ nodes to hold A rather than drawing the count binomially; f is therefore exact up to integer rounding. Positions are random except in the star tests, where the entry leaf and hub or controller are always among the A-holding nodes and the remaining $\text{round}(fN) - 2$ positions are random. This makes every star base informative about hub removal while retaining the exact completion count. Every canonical architecture has one external source, placed as the table says. Held-out canonical dimensions scale with N as stated below, so the active structure occupies the same share of the estate.

Because every canonical node holds X natively ($p_X = 1.0$), the completion fractions f_{XA} and f_A coincide across the canonical set, so the canonical law tests exercise the Authority gate alone. The class-specific completion fractions are distinguished only in the mixed architectures of Section 6.2, where p_X varies; the canonical transfer results should be read as demonstrating the gate shift for the Authority corner rather than across all classes.

The mesh grid is fixed before generation. Let $\bar{\pi}_{\text{conn}} = 0.12(0.55)(0.55)$, $\bar{T}_H = 1 - (1 - \bar{\pi}_{\text{conn}})^{50}$, and $d_{N,f}^* = 1/[(N - 1)f\bar{T}_H]$. Its twelve density levels are

$$d_j = \min\{0.95, d_{N,f}^* 2^{-2+5j/11}\}, \quad j = 0, \dots, 11.$$

They are evaluated separately for every N and f . If floating-point clipping creates a duplicate, the duplicate is retained as a separately seeded replicate at the pre-specified boundary rather than replaced after inspection.

Shape	Construction	Source	Varied	Architectures per sample
Mesh	the external source and estate nodes form one Erdos-Renyi mesh; every possible source-estate and estate-estate Connection is drawn at density d	the source is a vertex of the mesh and may have zero, one or many first internal touchpoints	the twelve fixed levels d_j above	$5 f \times 12 d \times 10 = 600$
Star, control-plane form	one directing plane whose controller is an estate node with n members	one connection from the source to one random member (entry at a leaf)	n in 10, 20, 40, 59; held-out 20, 40, 80, 119	$5 f \times 4 n \times 10 = 200$ bases
Star, connection form	one hub node with connections to n leaves	one connection from the source to one random leaf	n in 10, 20, 40, 59; held-out 20, 40, 80, 119	$5 f \times 4 n \times 10 = 200$ bases

Shape	Construction	Source	Varied	Architectures per sample
Star removal variants	for each base: intact; hub deleted with its relations; 1 random leaf deleted; 5 random leaves deleted (never the entry leaf); held-out deletions scale to 2 and 10 leaves	as the base	matched on the base	400 bases x 4 = 1,600
Star, source at the hub	the control-plane form with the source as the plane's controller (the CrowdStrike arrangement)	the source is the hub	n as above	5 f x 4 n x 10 = 200; exploratory only
Chain, channel form	Channel links 1 to 2, 2 to 3, and so on to depth 20; held-out depth 40	a Channel from the source to node 1	depth read off per node	5 f x 10 = 50
Chain, connection form	the same path built from Connections	a Connection from the source to node 1	as above	5 f x 10 = 50
Matched-count blocks	one node set and native draw per block; the same N minus 1 Connections arranged as a random mesh, as a hub with N minus 1 leaves, and as a path of N minus 1 links	one Connection from the source to the same first internal touchpoint in all three arrangements	shape only; 59 relations at N = 60 and 119 at N = 120	5 f x 20 blocks x 3 = 300

The matched-count blocks are the shape test. Class, scaled relation budget, node set, native corners, first internal touchpoint and run seeds are identical across the three arrangements, so any difference in spread is arrangement alone.

6.2 Mixed architectures (H4, H6, exploratory)

Factors are drawn by Latin hypercube, 400 architectures per sample. Connection and Channel classes are always present at their sampled densities; a Control Plane class is absent exactly when its sampled count is zero. Control-plane sources are dedicated nodes with native I, X and A, governed inside (estate) or outside (vendor) the boundary. Continuous factors map linearly from their Latin-hypercube coordinate to the stated closed interval. An integer factor on a to b maps coordinate u to $a + \min\{b - a, \lfloor (b - a + 1)u \rfloor\}$. Counts, rounded memberships and boundary assignments are therefore determined before any outcome is generated.

Factor	Development and replication	Held-out
N estate nodes	40 to 120	120 to 200
p_I, p_X, p_A (independent)	0.20 to 0.90	as development
co-location rho (Gaussian copula)	0.0 to 0.8	as development
connection density	0.01 to 0.15	0.15 to 0.30

Factor	Development and replication	Held-out
channel density (ordered pairs)	0.005 to 0.05	0.05 to 0.10
conferring planes: background count, membership	0 to 3; 5% to 50% of nodes	as development
directing planes: background count, fan-out	0 to 3; 5% to 60% of nodes	0 to 3; 60% to 95%
share of plane sources outside the boundary	0.0 to 1.0	as development
bundling: P(channel with a connection), P(connection with a directing membership)	0.0 to 1.0	as development
source count	deterministic cycle 1, 2, 3 by architecture index	same cycle
relay reach share	20% to 100%	as development
registry fan-in	1 to 5	as development

The held-out sample draws N, densities and fan-out outside the development ranges; that is its only difference. Design factors are recorded but never used as predictors; only the static measures of Section 7 enter any model.

6.3 Sources: how a payload enters

Every architecture carries one or more external source nodes, outside the boundary, with native I, X and A. A run pins entry to one source, which is compromised at step 0 by assumption and delivers on every relation it supplies. Each receiver’s triangle then decides whether the payload executes there. Nothing else enters the estate: phantom rates are zero.

Source type	Relations into the estate	Real-world case	Frozen delivery rule
Relay	connections to a share of estate nodes (canonical: fixed as in 6.1; mixed: 20% to 100% of nodes)	a mail relay that every desktop is connected to	receiver must hold X and A
Update	controller of one directing plane	a management or update agent	plane owned at step 0; members must hold X and A
Registry	channels into 1 to 5 estate nodes	a package registry feeding builds	receiver must hold A

Mixed architectures use $1 + (a \bmod 3)$ sources for zero-based architecture index a , giving 134 one-source, 133 two-source and 133 three-source architectures in each 400-architecture sample. The sample seed fixes one base permutation of relay, update and registry; architecture a rotates it by $a \bmod 3$ and takes the first configured number of types. Assignment is therefore without replacement, every three-source architecture contains one of each type, and the starting type is balanced across architecture indices. Relay reach is a separate Latin-hypercube factor mapped to 20%–100% of estate nodes. Registry fan-in is a separate integer factor from 1 to 5. Each update source controls one dedicated directing plane whose fan-out uses the architecture’s directing-plane fan-out factor; these source planes are additional to the sampled background plane count and enter every static measure. Confirmatory runs allocate exactly the configured number of trials to each source rather than sampling sources within a fixed architecture budget. The pilot uses 100 trials per source and confirmatory sets use 1,000 per source unless the pre-generation TIME rule in Section 12 selects 500. The per-source p_i is therefore estimated at equal precision and pooled with equal weight per source. This is the model’s form of the delivery layer: the source carries, the estate executes. An external email relay illustrates Paper 1B’s governance rule: its cross-boundary Connection is a system-of-systems Interface relation (Concentration), while

every desktop it reaches still needs its own X and A. A node cannot pass anything on until it has executed; carriage by uncompromised estate nodes stays a declared revision (Section 17).

7. Measures, all static, from the supply table

Every measure is computed from the frozen RelationTable rows before any run, with the same definitions as `cemt_core.relations` (one row per condition supplied, connection rows symmetric, node-level self-supply as rows). Nothing about a run enters a measure.

7.1 Per node (the predictors of per-node probability)

Measure	Definition
completion	whether the node's triangle completes for entry (I, X, A) and for each delivery class (Section 5), and whether each corner is native or supplied
delivery density	number of relation rows that can deliver to the node, by class
gated distance	source-specific fewest internal compromise hops from a first touchpoint inside the system, counting a hop only into a node that completes for that hop's class; each first touchpoint has distance 0; membership of a directing plane is one hop from its reachable controller
ungated distance	the same source-specific hop count from the same first touchpoints, ignoring completion (topology only): the rival for H6
gated route redundancy	number of internally vertex-disjoint delivery paths from the source's first internal touchpoints to the node through class-complete nodes; the touchpoints connect to a super-source by separate capacity-one arcs
ungated route redundancy	the same source-specific path count ignoring completion
reachable hub membership	number of directing planes with a controller reachable from that source which the node belongs to, and the largest fan-out among them
boundary supply	which corners, if any, are supplied from outside the governance boundary (system-of-systems status under the Section 3.2 rule)
Cut	for each specified operation and corner, whether a supplying node lies on every available supply of that condition, tagged by relation class, exactly as Paper 1B equation (10) defines it

7.2 Per architecture

Measure	Definition	Shape it captures
completion fractions	<code>f_IXA</code> , <code>f_XA</code> , <code>f_A</code> ; native-only and with supply	the gate
complete-triangle count	number of nodes complete for delivery	the count for H5
redundancy	mean number of suppliers per supplied corner	mesh
hub concentration	largest share of complete triangles that one supplier completes; Freeman centralisation of the delivery graph	star
chain depth	longest run of single-supplier links (a sequence of Cuts)	chain
Cut density and Fan-out by class	as computed from the relation table in 3A	1B properties
boundary share	share of supplied corners with a source outside the boundary	system of systems

Measure	Definition	Shape it captures
topology-only set	mean and maximum degree, directed betweenness, source-specific reachable-set size from the first internal touchpoints, strongly connected components, mean shortest path, on the directed union of all relations with class labels removed	rivals

All graph algorithms are implemented in the file: breadth-first search for distances, Brandes for betweenness, Edmonds-Karp after node-splitting for internally vertex-disjoint paths, and Tarjan for strongly connected components. For max-flow, every internal vertex v is replaced by $v_{in} \rightarrow v_{out}$ with capacity one; delivery edges have capacity one; the super-source and destination are not capacity-limited. This prevents edge-disjoint paths that reuse an internal vertex from being counted as redundant. N never exceeds 200 plus external sources, so exact computation is affordable.

8. Outcomes and run records

The primary outcome is the probability that each node is compromised given delivery from a source. The source is compromised by assumption, so every run counts and the outcome isolates propagation (the ESRB amplification phase) from the node-level entry gate that Paper 1A already covers.

Outcome	Definition	Used by
p_i	share of runs from a source in which node i is ever compromised; reported per source and pooled with equal weight per source	H3, H4, H6; every per-node prediction
Y	mean fraction of estate nodes ever compromised (the mean of p_i)	H1, H2, H5
first-hop rate	share of runs in which at least one estate node is compromised	descriptive: the share of deliveries that execute anywhere
P_collapse	share of runs meeting the frozen sustained-collapse rule ($x_{true} \geq 0.30$ for 3 steps)	secondary, for continuity with 3A and v8
impact reached	mean total node impact compromised, using the model's per-node impact values	descriptive only (equal-value assumption)
variance split	share of within-architecture variance of the per-run fraction explained by the source, where there are several	exploratory

Per run the driver records the architecture id, run seed, source, every estate node ever compromised with the step it was first compromised, the frozen collapse flag and the number of steps. Runs are written to runs.csv.gz. The per-node table (one row per node per architecture and source, with its Section 7.1 measures and p_i) is the analysis dataset for H4 and H6.

With 1,000 runs per source, every per-source p_i has maximum binomial standard error $\sqrt{0.25/1000} = 0.0158$. Canonical architectures have one source; mixed architectures receive the same number of runs separately for

each of their one to three sources. Y is calculated once per run and then averaged; node outcomes within a run are correlated, so its precision is estimated from the variance of the run-level compromised fraction rather than by treating nodes as independent. Pilot runs use 100 runs per source.

9. Analytic predictions

Every hypothesis has a prediction written before the run, from the frozen rates, the realised gates and the 50-step horizon. Predictions are computed by the analysis code from each architecture's supply table and gate draws, so the comparison is per architecture, not to a textbook constant.

9.1 Per-step delivery probabilities in v0.8 (Time and Adaptation off, exec_modifier = 1)

$$\begin{aligned}\pi_{\text{conn}}(j) &= \beta_{\text{conn}} x_j a_j T, \\ \pi_{\text{chan}}(j) &= \left(1 - \frac{1}{1 + \delta}\right) a_j T, \\ \pi_{\text{dir}}(j) &= \beta_{\text{conn}} \min(1, b) x_j a_j T.\end{aligned}$$

with β_{conn} 0.12, δ 2.0, b 2.0, T 1.0, and x_j, a_j the realised gates. A hop attempted over h remaining steps succeeds with probability $1 - (1 - \pi)^h$ to the power h . A directing plane with m compromised members is taken over in a step with probability $1 - 0.92^m$ to the power m , and at once if its controller is compromised.

9.2 The ceiling (theorem, verified as an implementation check)

For source s , let E_s be the union of estate nodes that complete for at least one delivery class on an ungated path from s . Then

$$Y_s \leq \frac{|E_s|}{N} \quad \text{and} \quad p_{is} = 0 \text{ for every } i \notin E_s.$$

In a canonical architecture containing one delivery class this reduces to $Y \leq f_{\text{class}}$. The union ceiling is required in mixed architectures because compromise through several classes may exceed each class-specific completion fraction separately without violating the gate.

9.3 Chain

For node k on a chain, number the first touchpoint inside the system as 0. With every node between that touchpoint and k complete:

$$p_k = \prod_{j=0}^k \mathbb{P}[\text{node } j \text{ completes}] \cdot \Pr \left[\sum_{j=0}^k W_j \leq H \right], \quad W_j \sim \text{Geometric}(\pi_j).$$

Here W_0 is delivery from the external source to the first internal touchpoint and later terms are internal compromise hops. The second factor is computed exactly by dynamic programming over steps. Predicted behaviour: p_k is non-increasing in k , falls geometrically with the number of completed hops where the horizon is not binding, and falls faster where it is.

9.4 Mesh (Erdos-Renyi connections; asymptotic, tested as ordering and scaling)

The external source is a vertex of the mesh and is governed by the same density draw as the other vertices. It may therefore have zero, one or many first touchpoints; seeding is the ordinary first step of the realised mesh, not a separately fitted process. With mean estate degree $c = d(N - 1)$, site occupation f_{XA} and horizon transmission $T_H = 1 - (1 - \pi_{\text{conn}})^H$ to the power H , the asymptotic occupied-component prediction is:

$$d^* = \frac{1}{(N - 1)f_{XA}T_H}, \quad S = 1 - e^{-cf_{XA}T_H S}, \quad E[Y] \approx f_{XA}S^2.$$

Here $f_{XA}S$ is the asymptotic estate fraction in the occupied giant component conditional on reaching it, while the source reaches that component with probability approximately S . Their product is the unconditional mean over the pre-generated Erdos-Renyi architectures used by H1. No additional fitted seeding parameter is introduced.

The equation gives the bulk scaling law; the realised source degree and first-touchpoint count are reported as diagnostics, not used to condition away architectures in which the source misses the giant component. Predicted behaviour: Y is near zero below d^* , rises steeply through it, and approaches the reachable completion ceiling f_XA at high density. The midpoint density, where unconditional mean Y reaches half its ceiling, remains proportional to 1 over f_XA , so midpoint times f_XA is approximately constant across the f levels, and halves when N doubles.

9.5 Star (removal)

With the hub deleted, a successfully compromised first internal touchpoint can reach nothing else. If its first-hop execution probability is p_0 , the expected estate fraction is p_0/N . With k of n leaves deleted, Y falls by about k over n of its intact value.

$$\Delta Y_{\text{hub}} \approx Y_{\text{intact}} - \frac{p_0}{N}, \quad \Delta Y_{\text{leaves}}(k) \approx \frac{k}{n} Y_{\text{intact}}.$$

Predicted behaviour: deleting the hub costs more than deleting five leaves at every f and every n .

9.6 Position (mixed architectures; functional form for the regression)

For a node at gated distance k from the first internal touchpoint, with r vertex-disjoint routes, first-hop execution probability p_0 , and effective internal per-hop probability q :

$$p_i \approx 1 - (1 - p_0 q^k)^r.$$

Predicted behaviour: $\log p_i$ falls linearly in gated distance for $r = 1$ and the fall is shallower as r rises. A node that is a member of a directing plane whose controller is reachable is assigned the corresponding control-plane edge when distance and redundancy are computed; hub membership is therefore represented inside the two structural predictors rather than entered again as an independent effect. The number of internal hops after which further distance adds little follows from q : at $q = 0.3$ the third internal hop contributes under 0.03 before the first-hop factor.

9.7 Shape at matched count

With the same N minus 1 Connections, node set, first internal touchpoint and gates, the imported theory predicts the chain gives the lowest Y (one narrow path, depth N minus 1), and the star and mesh both exceed it; the ordering of star against mesh depends on the entry position and is reported without a prediction.

10. Hypotheses and decision rules

A hypothesis is supported only if every one of its criteria holds on each replication seed separately (4201 and 4202) and on the held-out sample (4301). All confidence intervals are 95% percentile bootstrap intervals over architectures (Section 13). The hypotheses are pre-specified and reported separately; the paper makes no omnibus familywise claim. Within a multi-criterion hypothesis every listed criterion must hold, and H3 uses a simultaneous rather than pointwise prediction band.

H	Law	Supported if
H1 mesh	percolation threshold, gated by f_XA	(a) at every f level, Y rises from below 0.10 f to above 0.50 f across the density grid, and a four-parameter logistic fit of Y on $\log d$ has a slope whose CI excludes 0; (b) the fitted midpoint density falls as f rises: Spearman between f and midpoint over the five levels is negative with bootstrap CI excluding 0; (c) midpoint times f_XA has a coefficient of variation below 0.25 across f levels, provided the pilot confirms this threshold is feasible against the midpoint sampling variance (Section 12); (d) the plateau at the highest density is within 0.05 of f_XA
H2 star	hub fragility	for both star forms, at every f and every scaled n : ΔY_{hub} minus $\Delta Y_{\text{leaves}}(k)$ has CI lower bound above 0, with $k = 5$ at $N = 60$ and $k = 10$ at $N = 120$

H	Law	Supported if
H3 chain	decay with depth	for both chain forms, at every f: (a) the maximum positive adjacent-depth difference in the observed aggregate profile is no greater than the simultaneous 95% critical value generated under the Section 9.3 model; (b) the complete observed depth profile lies inside the simultaneous 95% prediction band calibrated by the maximum standardised deviation under that model; (c) at $f = 1$ the maximum absolute deviation from prediction is at most 0.05, isolating the horizon term from the gate
H4 position	gated distance law (mixed architectures)	in a binomial regression of p_i on gated distance, log gated route redundancy and their interaction (Section 13): (a) the distance coefficient is negative, CI excluding 0; (b) the redundancy coefficient is positive, CI excluding 0; (c) the distance by redundancy interaction is positive, showing that redundancy weakens distance decay, CI excluding 0. The model is fitted independently on replication 4201, replication 4202 and held-out 4301, with transformations and the estimation procedure fixed from development, and all three fits must pass
H5 shape	shape beyond count (matched-count blocks)	(a) chain below star and chain below mesh: paired mean difference in Y with CI lower bound above 0 and at least 0.05; (b) star against mesh reported, not tested
H6 gate	gated position beats topology	among nodes that complete for at least one delivery class available from the source and are reachable in the ungated graph, concordance of the pre-specified position form using gated distance and gated redundancy exceeds concordance of the same form using ungated distance and ungated redundancy by at least 0.02, paired bootstrap CI lower bound above 0. The primary rival computes ungated distance and redundancy on the subgraph induced by the eligible nodes only; the whole-graph ungated rival is reported as a secondary, weaker comparison. Directing-plane edges enter both pairs of distance and redundancy measures, and both models are fitted on development only and scored unchanged

H1 to H3 are instrument validation: they test whether v0.8 reproduces the imported laws and whether the gate shifts each by its completion fraction. The only prediction among them that is not close to built in is H1(c), the scaling of the mesh midpoint with 1 over f_XA . The paper’s novel content is H4 to H6, where it says something the imported results do not: that position must be measured through the triangle, not the network alone. H6 holds completion eligibility and hub information constant between models, so its contrast is the arrangement of completed paths rather than the built-in fact that incomplete nodes cannot execute.

Reporting a partial pass. A hypothesis that meets every criterion on both replications but fails in a bounded set of held-out cells is reported as *supported under replication, not generalised*, with the failing cells and their margins named. A hypothesis that fails a criterion on a replication seed is reported as *not supported*, again with the failing cells named. No threshold, criterion or model is altered to convert a fail into a pass; this rule is fixed before the confirmatory run so a near-miss is narrated, not rescued.

11. Implementation checks that abort the run

These are not hypotheses. Each is true of a correct implementation of v0.8 by construction, and a failure means the driver, the measures or the embedded core are wrong. A failure stops the run before any hypothesis is evaluated.

Check	Condition	Where
IC1 freeze	embedded core hash equals digest 97dbae47	at load

Check	Condition	Where
IC2 ceiling	for every architecture, source and run, every compromised node belongs to the source-specific eligible union E_s , and $Y_s \leq E_s /N$; in a single-class canonical architecture this also requires $Y \leq f_{\text{class}}$	every run
IC3 structural exactness	in deterministic mode at $f = 1$, each chain is reached to its configured end (depth 20, or 40 held-out), the intact star reaches every member, and the hub-deleted star reaches only the first internal touchpoint	canonical architectures, pilot and confirmatory
IC4 exact canonical gate	the number of A-holding nodes equals $\text{round}(fN)$, and realised f_A differs from its target only by integer rounding	at build
IC5 matched blocks	the three arrangements of a matched-count block share the node set, native corners, gate draws and relation count exactly	at build
IC6 measures	for the same source and node, finite gated distance is greater than or equal to ungated distance; a node unreachable ungated is also unreachable gated; gated redundancy is no greater than ungated redundancy; redundancy is defined as 1 at a first touchpoint and is at least 1 for every other node with finite distance; a two-path fixture sharing one internal vertex returns redundancy 1 after node-splitting; Cut values reproduce Paper 1B equation (10) on single-supply, alternative-supply and shared-upstream fixtures	at build
IC7 samples	no pilot seed appears in a confirmatory sample; each replication seed produces its own complete sample	at generate
IC8 validate guard	<code>cemt_core.ScenarioSpec.validate</code> is asserted never to be called on any built spec, so the latent Relation-Class.DEPENDENCY reference (Section 17) cannot be tripped by a later edit	at build

The check that replaced the v9 extension-ratio abort is IC2: it tests the theorem the paper relies on, and cannot fail on a valid baseline.

12. Samples, seeds and budget

There are four sample roles and five generated sets: one pilot, one development, two independent replications and one held-out set. Verdicts require every criterion to hold on both replication seeds and on the held-out sample.

Sample	Seed	N (canonical / mixed)	Role
Pilot	231 (with 543 and 311 as further pilot seeds)	60 / 40 to 120	design checks, broad screen, no verdicts; 100 runs per source
Development	4101	60 / 40 to 120	fixes predictor transformations and estimation procedure, fits the H6 models and supplies logistic-fit starting values; no verdicts
Replication	4201 and 4202	60 / 40 to 120	verdicts, each seed separately
Held-out	4301	120 / 120 to 200	verdicts beyond the development ranges

Per generated set: 600 mesh, 1,600 star (400 bases with four removal variants), 200 hub-source star, 100 chain, 300 matched-count and 400 mixed architectures, 3,200 in all. Canonical architectures use 1,000 runs each and mixed architectures use 1,000 runs for each source; the primary pilot uses 100 on the same basis. The deterministic 1, 2, 3 source cycle gives 799 mixed architecture-source units and 3,599 run-budget units per generated set. These sizes are the starting point: the pilot reports the architecture-level variance for H5 and H6, and sizes are raised as required before the specification is frozen (decision of 20 Sept 2026). Architecture seeds are derived from the sample seed, the family, the cell and the index, so every architecture is reproducible from the run record alone.

Budget: at $N = 60$ with Time and Adaptation off, one 1,000-run architecture-source unit takes about 2 to 4 seconds of one core in the v9 pilot code (a 50-run P11 architecture at $N = 64$ took 0.1 s). The four full confirmatory sets contain 12,800 distinct architectures and 14,396 run-budget units, requiring about 8.0 to 16.0 core-hours at that benchmark before allowing for the larger held-out architectures. The primary pilot runs the same 3,599 units at one tenth of the confirmatory Monte Carlo budget. A TIME mode times 50 architecture-source units at 1,000 runs, stratified across canonical and mixed families and including held-out sizes, before confirmatory generation and writes the estimate to the run record. If its projected wall time exceeds 8 hours on the available machine, runs per source drop to 500 for every family; that decision is recorded before any confirmatory architecture is generated.

Precision and power: each p_i has maximum binomial standard error 0.0158 at 1,000 runs and 0.0224 if the frozen budget is reduced to 500. Y is a run-level fraction with correlated node outcomes, so no gain of $\sqrt{N}$ is assumed; its standard error is estimated from the pilot distribution of run-level Y . For H5, the smallest effect the paired test must detect is 0.05 in Y ; with 100 matched blocks per set and between-block standard deviation below 0.15 (pilot to confirm), power exceeds 0.9. For H6, the effective resampling unit is the 400 mixed architectures, not their constituent nodes; the pilot supplies the architecture-level variance of the concordance difference so the mixed set can be enlarged before freezing if required. For H1 to H3, the pilot reports the sampling variance of each fitted quantity that a criterion tests: in particular the fitted mesh midpoint per f level, so that H1(c)'s coefficient-of-variation threshold is confirmed achievable against that variance before the specification is frozen, and the chain depth-profile deviation for H3. Any criterion the pilot shows to be infeasible on noise alone is loosened to a pre-recorded value before freezing, never after the confirmatory run.

13. Statistical procedure

Every estimator, transformation and resampling rule is fixed here; the analysis code implements these and nothing else.

Element	Rule
Unit of analysis	the architecture for H1, H2, H3, H5; the node within architecture for H4 and H6

Element	Rule
Bootstrap	2,000 resamples, seed 20260920; preserve every designed cell by resampling the ten architectures within each H1 f, d cell, the ten bases within each H2 f, n , form cell, the ten chains within each H3 f , form cell, and matched blocks within f for H5; H4 and H6 resample architectures with all sources and nodes attached; paired differences use the same resample for both arms; 95% percentile intervals
Matched pairs (H2, H5)	differences computed within base or block, then bootstrapped over bases or blocks; never over runs
Logistic fit (H1)	four-parameter logistic of Y on $\log d$ per f level, floor fixed at 0, ceiling free, fitted by least squares (<code>scipy.optimize</code>) with starting values from the development sample; the midpoint is the fitted location parameter
Prediction band and monotonicity (H3)	per chain form, f level and evaluation sample, simulate 2,000 complete batches from the Section 9.3 geometric waiting-time model conditional on the actual ten chains, including each chain's realised A placement, x_prob , a_prob and aggregation across chains; preserve dependence across depths; the 95th percentile of the batch maximum absolute standardised deviation gives the simultaneous band, and the 95th percentile of the batch maximum positive adjacent-depth difference gives H3(a)'s monotonicity critical value; a depth with zero predicted variance is compared on the unstandardised scale and any non-zero deviation is outside the band
Binomial regression (H4)	events = runs in which the node was compromised, trials = runs from that source; logit link; predictors are gated distance, $\log(1 + \text{gated route redundancy})$, their interaction and a gated-unreachable indicator; directing-plane edges are already included in distance and redundancy and membership is not entered again; transformations are standardised using development means and standard deviations. The model is estimated by Firth penalised likelihood, which corrects small-sample bias and separation without shrinking the target interaction coefficient toward zero; if an additional ridge is required for numerical stability, its value is selected on the development sample by 5-fold cross-validation and then frozen, with the value and the cross-validation curve recorded in the run record. The frozen procedure is then applied separately on replication 4201, replication 4202 and held-out 4301, with architecture-clustered bootstrap intervals in each sample, and H4(c) requires the interaction sign and interval to hold in all three fits
Fair rival (H6)	evaluation is restricted to source-node rows that complete for at least one delivery class available from that source and are reachable in the ungated graph; the gated and ungated models use the same functional form, training rows and fitting procedure, differing only in gated versus ungated distance and redundancy; both include directing-plane edges through their respective distance and redundancy measures and neither has a separate membership term; both are fitted once on development and scored unchanged on each evaluation sample. The primary ungated rival computes distance and redundancy on the subgraph induced by the eligible nodes only, so the contrast is the arrangement of completed paths and not the built-in inability of ineligible nodes to execute; the whole-graph ungated rival is reported as a secondary comparison
Concordance (H6)	share of eligible node pairs within the same architecture and source whose predicted order matches the order of p_i ; pairs with equal p_i excluded; score ties count one half; each bootstrap draw independently resamples development architectures for refitting and evaluation architectures for scoring, keeping all sources and nodes of an architecture together and using the same draws for both models
Spearman (H1b)	<code>scipy.stats.spearmanr</code> ; cell-stratified bootstrap as above
Missing values	a node with infinite gated distance has $p_i = 0$ because it is outside the source-specific eligible union, unreachable from that source through completed paths, or both; for H4 its numeric distance is set to the architecture-and-source maximum finite distance plus 1 and the gated-unreachable indicator is set to 1; H6 uses only the eligible rows defined above and encodes any gated-infinite distance as the eligible set's maximum finite gated distance plus 1 without adding a model-specific indicator; every exclusion category is reported

No hypothesis, criterion, threshold or model is changed after the confirmatory run starts. Anything learned from the confirmatory results is reported under Section 15 as exploratory.

14. Docker correspondence

The two-world method carries over from 3A, and the Docker test compares the same construct in both worlds: the reach set in deterministic mode. This answers the v9 review’s objection that a static measure was being compared with a dynamic observation.

- Docker-scale counterparts: nine predetermined architectures generated from the development seed 4101 at $f = 0.6$ and $N = 30$, so that no verdict seed (4201, 4202, 4301) is spent on an instrument check. They are not described as a subsample of the $N = 60$ replication architectures. The three meshes use density levels nearest $0.5d^*$, d^* and $2d^*$; the three stars are a control-plane star with $n = 20$, its matched hub-deleted negative control, and a Connection star with $n = 20$; the three chains are a Connection chain of depth 12, a cumulative Channel chain of depth 12, and its matched negative control with the sixth node incomplete while the total A count is preserved by moving A to a node outside the chain. Their identifiers and specifications are written to `docker_h2.csv` before any container runs.
- Rendering: the v0.8 Docker renderer, 20 seeds each, round-indexed as in 3A; entry pinned to the same node in both worlds.
- Correspondence claim: the container-observed reach set equals the model’s deterministic reach set exactly, for every architecture and seed. The hub-deleted star is the negative control. One mismatch is reported as a failure with its diagnosis, as c18 was in 3A.
- Status: this is an implementation-level correspondence check, exempt from the held-out requirement, and stated as such. It is not independent validation against real systems.
- Capacity is preflighted before the specification is frozen. If Docker Desktop cannot host all nine architectures, their common N or shape sizes are reduced and this section is amended before freezing. After freezing, failure to run any required architecture is a correspondence-check failure, not an exclusion.

15. Exploratory outputs

Reported descriptively, with no verdicts, and labelled exploratory in the paper.

Output	Content	Why
v8 reanalysis	on the v8 validation files: scenarios grouped by supplied-capability quintile, mean collapse rate against connection density; does spread plateau at the level the overlap sets?	prior evidence for the ceiling from data already seen; must be labelled post hoc
Star entry asymmetry	hub-pinned against leaf-pinned entry: Y and steps to reach half the members	time-to-spread is a 4B quantity; recorded here for the 4B baseline
Star against mesh at matched count	the ordering left unpredicted in H5	informs the 4B and 4C lever predictions
Boundary supply	p_i for nodes whose X or A is supplied from outside the boundary against nodes supplied inside, at matched gated distance	descriptive for the system-of-systems paper; largely by construction in v0.8
Diminishing hops	the effective q per hop and the depth at which predicted p_k falls below 0.05, per f	Paul’s two-to-three-hop intuition, stated from first principles

Output	Content	Why
Value-weighted impact	impact reached against Y, and the per-node p_i of the highest-impact decile	the equal-value assumption made visible; sets up placement work
Broad screen (pilot only)	every Section 7 measure ranked by single-measure concordance with p_i and with Y	the reveal layer from the original Paper 4
Variance split	share of within-architecture variance explained by the entry node, by family	structural against random
Realised factor correlations	the correlation matrix of design factors per sample	the Latin hypercube caveat
Pair against singles	not applicable in this design; retained only if mixed architectures are later masked by class	recorded so the omission is deliberate

16. Output files and run record

One file per table so the manuscript can paste them inline. All go to the run directory named with timestamp, mode and version.

File	Contents
run_record.json	Section 2 contents: code and specification versions and hashes, core digest, hash check, Python and library versions, RNG, every range, seed, family and cell definition, timestamps, TIME-mode estimate, the H4 cross-validation curve and frozen ridge value if any, exclusions
architectures.csv	one row per architecture: sample, family, shape, cell, seed, design factors, every Section 7.2 measure, realised gates summary
nodes.csv.gz	one row per node, architecture and source: every Section 7.1 measure, realised x_prob and a_prob, p_i, events and trials
runs.csv.gz	one row per run: Section 8 fields
outcomes.csv	one row per architecture and source, plus the equal-source pooled row: Y, first-hop rate, P_collapse, impact reached and variance split
predictions.csv	per architecture or node the Section 9 prediction beside the observation
verdicts.json	every criterion of every hypothesis with its estimate, interval and pass or fail, per sample and overall, including the partial-pass classification of Section 10
verdicts.txt	the console summary as printed
docker_h2.csv and docker_h2_specs/	the nine predetermined Docker-scale counterparts, static reach sets, and the scenario files to render
exploratory.csv	Section 15 outputs
specs/	every architecture as a scenario file, for 4B and 4C and for figure generation
figures/	the three generated figures and the H1 to H3 law plots

Modes: PILOT (pilot seeds, reduced budget, broad screen, no verdicts), TIME (budget estimate), ALL (generate then analyse), GENERATE (resumable), ANALYSE (existing run). AMEND is not provided: a change to the design after the confirmatory run is a new version and a new run.

17. Limitations and declared revision candidates

The paper states each of these. None blocks the Structure pillar on v0.8; each is a candidate for a declared v0.9 revision when 4B or 4C needs it.

Limitation of v0.8	Effect on this paper	Revision candidate
No carriers: only a compromised node passes anything on	delivery through uncompromised nodes (mail relay, registry, load balancer) is absent; distance is compromise hops; the position law understates arrival where real carriers exist	carriage independent of compromise, per relation
Directing plane supplies I only; the member's X and A must hold	completion from zero by a directing plane cannot occur; the CrowdStrike arrangement is represented as the agent holding X and A locally	directing supply of X or A as a spec option
Compromise of a conferring issuer confers nothing	the Golden SAML step of SolarWinds cannot occur	issuer compromise confers A on members
Push and pull are not distinguished	where the Interface must exist is not represented	direction attribute on directing planes
The v0.8 Channel delivery operation bundles admission with traversal	measures and predictions use the frozen bundled delivery rule, while Paper 1B continues to classify Channel itself on the X row conditional on admission; the receiver must hold A	expose admission and route as separate implementation stages in a later core revision
Plane ownership persists after the controller is remediated	irrelevant with Adaptation off; matters for 4C	ownership tied to controller state
<code>ScenarioSpec.validate</code> references <code>RelationClass.DEPENDENCY</code> , which does not exist	<code>validate</code> is not called (guarded by IC8); latent crash	fix at refreeze
Beta gate draws cannot be made constant	predictions use realised gates	none needed

The paper also states: results hold on the frozen model only; field calibration is deferred; the equal-value assumption; and that H6 is partly favoured by construction, which the induced-subgraph rival of Section 13 is designed to limit.

18. Terminology

- The conditions are Interface, Execution Pathway and Authority, written I, X and A. The acronym IAE is not used.
- The node-level model is the Cyber Triangle (Paper 1A). A triangle is complete when all three corners hold, natively or by supply.
- Supply and conferral are Paper 1B's terms and the only ones used for a corner obtained through a relation. Inherit is not used in the paper.
- Relations that cross a governance boundary are system-of-systems relations. Systemic is reserved for consequences; the systemic event is what a completed chain of triangles can cause.
- Delivery is what the system and system-of-systems layers do; execution is what a node does. A carrier is a node that delivers without executing (not represented in v0.8).
- Mesh, hub-and-spoke (star) and chain are shapes of triangle fabric, measured as redundancy, hub concentration and chain depth.
- Gated distance is source-specific internal compromise hops from the first touchpoint through complete triangles; ungated distance uses the same source and touchpoint while ignoring completion.
- Pre-specified frozen evaluation set, not pre-registered, following 3A's usage.

19. Decisions for Paul before freezing

Settled on 20 Sept 2026, unless marked open.

- ☒ Configuration: Time and Adaptation off (M_S).
- ☒ Entry: an external source, compromised by assumption, delivering on all its relations; per-node probability given delivery is primary.
- ☒ Sample sizes: start from Section 12, run the pilot, raise as required, then freeze.
- ☒ VERSION 12 for code, specification and tag; the unshipped v9 to v11 drafts are superseded.
- ☒ Paper 2 resubmission states Structure as completion and arrangement; Paper 1B v2.1 remains authoritative for relation rows and governance levels; Paper 4A separately records the bundled delivery behaviour of cent_core v0.8 without redefining Channel.
- ☒ H4 estimation: Firth penalised likelihood, with any stability ridge chosen by cross-validation on development and frozen, rather than a hand-set penalty (decision of 20 Sept 2026).
- ☒ H6 rival: the primary ungated rival is computed on the eligible-node induced subgraph; the whole-graph rival is secondary (decision of 20 Sept 2026).
- ☒ Docker counterparts generated from the development seed 4101, not a verdict seed.
- ☐ H5 direction: chain lowest predicted; star against mesh unpredicted (default) or predicted.
- ☐ Docker-scale counterparts: nine predetermined architectures at $N = 30$; confirm all nine run in Docker Desktop before freezing.
- ☐ A second held-out seed: add one so that generalisation beyond the development ranges is itself replicated, or accept the single 4301 draw and state it as a limitation. This is a compute-budget decision.
- ☐ Paper 3A v1.1 and Paper 2 roadmap sentences updated to Structure (4A), Time (4B), Adaptation (4C) before 3A is submitted.
- ☐ Thresholds (the former 4B): fold into each pillar as its collapse point, or drop.
- ☐ The exploratory v8 reanalysis: include labelled post hoc, or hold back.

The specification is frozen once the pilot has passed every implementation check, the H1 to H3 criterion thresholds are confirmed feasible against the pilot sampling variances, and the sample sizes are confirmed. The code is then written to it, committed and tagged v12, and the confirmatory run starts.